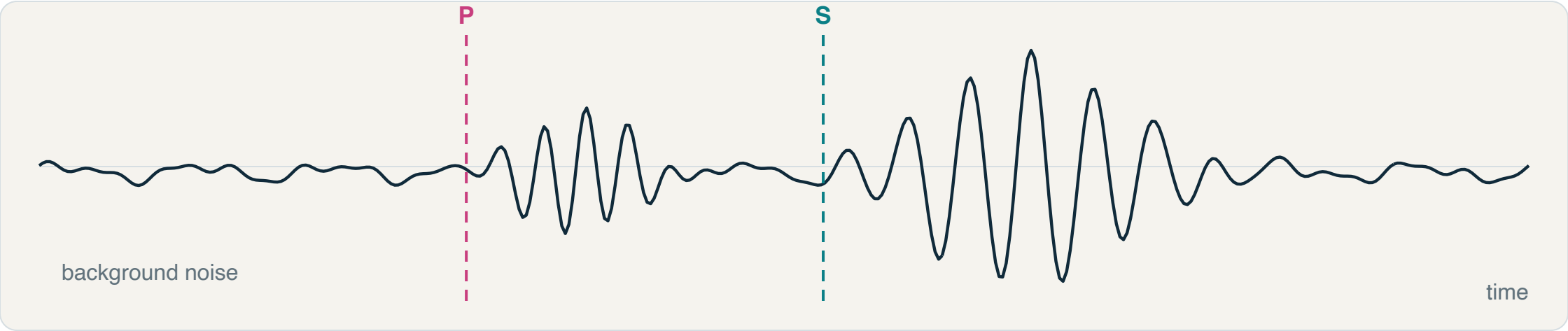
The arrival-picking problem

A picker converts a changing waveform into reproducible P and S arrival times



Onsets emerge gradually

The first visible wiggle can depend on zoom and judgement.

Noise can look like an arrival

A repeatable rule must separate signal from background.

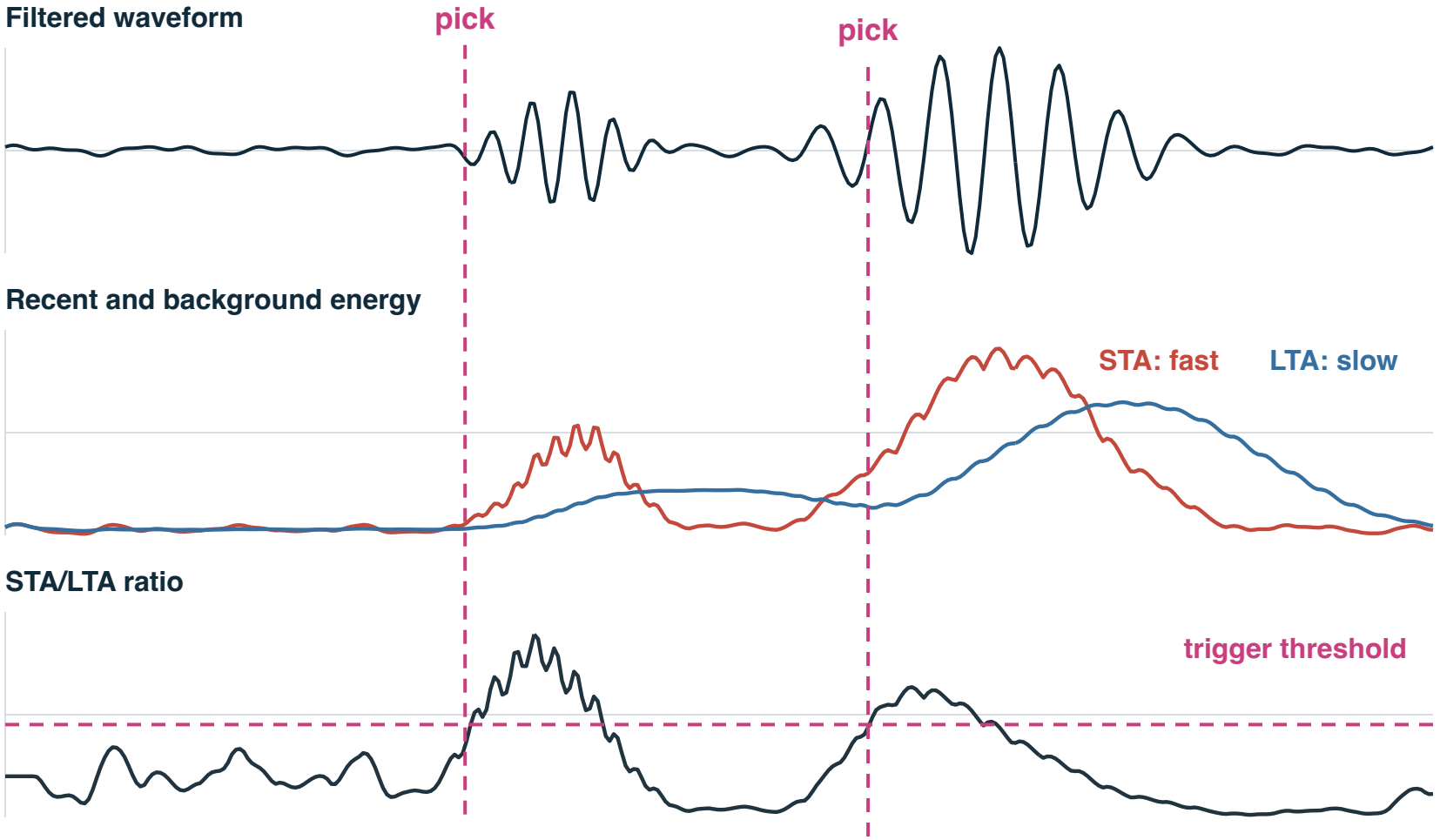
Hundreds of traces need consistency

Automatic picks still require visual quality control.

The algorithm proposes a time. The analyst judges whether that time follows the physical phase.

STA/LTA: recent energy versus background

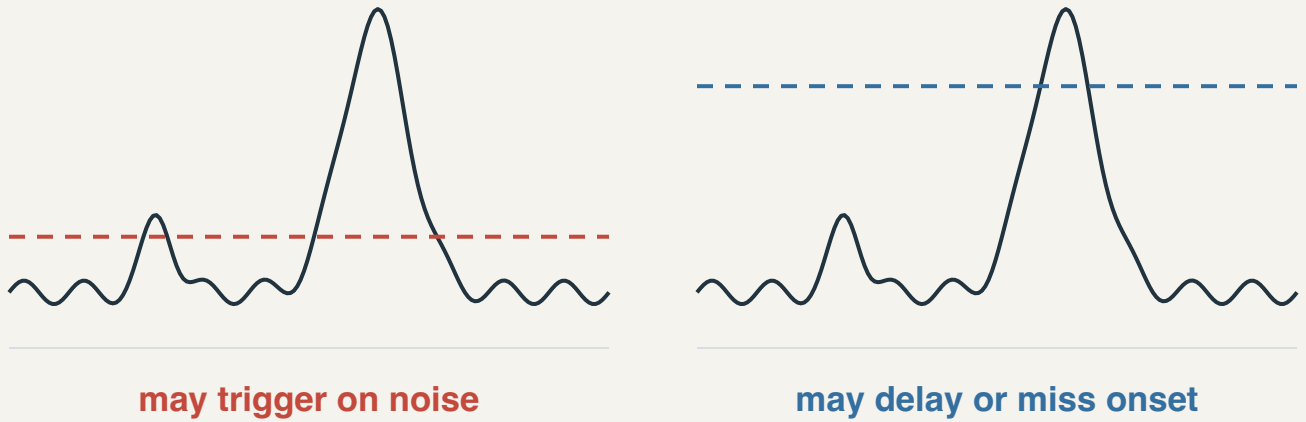
The short window responds quickly; the long window represents the preceding noise level



1. **Short average**
   follows the newest energy
2. **Long average**
   tracks the background
3. **Ratio rises**
   when the signal changes suddenly
4. **First crossing**
   becomes the proposed arrival

The current routine uses causal averages. Each decision uses the present and earlier samples.

STA/LTA parameters control sensitivity and timing



A good setting separates the true onset from background fluctuations across many traces.

1. **STA window**
   Shorter responds faster, but follows isolated spikes more strongly.

2. **LTA window**
   Longer gives a steadier background estimate, but adapts slowly.

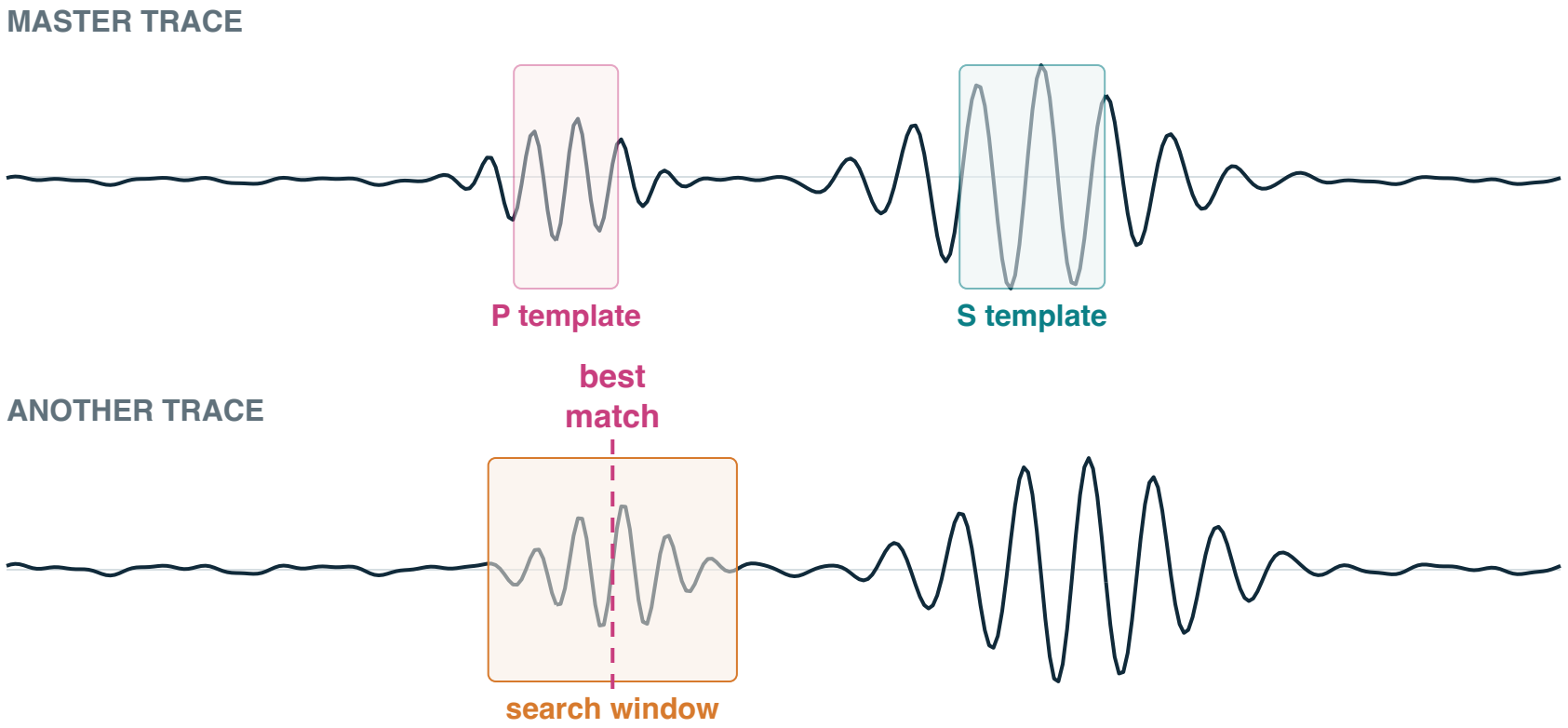
3. **Threshold**
   Lower increases sensitivity. Higher demands a clearer onset.

4. **P before S**
   The routine finds P first, then searches for S after a minimum delay.

GUI defaults: STA 0.5 μs LTA 15 μs
P threshold 3.5 S threshold 2.5

Cross-correlation: matching a familiar waveform shape

A master trace supplies separate P and S templates that are matched across the experiment



1. **Choose a master**
   Use a clear, representative trace.
2. **Mark P and S**
   The clicks define each reference arrival.
3. **Cut two templates**
   Keep the local shape around each phase.
4. **Search nearby**
   The strongest normalized match gives the pick.
5. **Reject weak matches**
   A minimum coefficient filters uncertain picks.

Defaults: P ±1.0 μs S ±1.5 μs
search ±5 μs accept CC ≥ 0.50

STA/LTA and cross-correlation

The methods provide complementary evidence. Disagreement identifies traces that need inspection

	STA/LTA	Cross-correlation
Core question	When does energy rise above background?	Where does a known phase shape match best?
Main strength	Can detect a new onset without a master trace	Tracks small arrival shifts consistently
Sensitive to	Noise bursts and threshold choice	Master quality and changes in waveform shape
Quality measure	Threshold crossing and visual consistency	Correlation coefficient and visual consistency

Validation workflow

1. **Run both pickers**
2. **Inspect the stacked traces**
3. **Locate disagreements or trends**
4. **Correct doubtful picks manually**

Learning check: Which method is most likely to trigger early on a noise spike? Which method may struggle if damage changes the waveform shape?